

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 2) - Single Correct

A mosquito is moving with a velocity $\vec{v} = 0.5t^2\hat{i} + 3t\hat{j} + 9\hat{k}$ m/s and accelerating in uniform conditions.

What will be the direction of mosquito after 2 s ?

(1) $\tan^{-1}\left(\frac{2}{3}\right)$ from x -axis

(2) $\tan^{-1}\left(\frac{\sqrt{85}}{6}\right)$ from y -axis

(3) $\tan^{-1}\left(\frac{5}{2}\right)$ from y -axis

(4) $\tan^{-1}\left(\frac{5}{2}\right)$ from x -axis

Q2: 18 March (Shift 1) - Numerical

A person is swimming with a speed of 10 m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is 'x' m/s. The value of 'x' to the nearest integer is _____

Answer Key

Q1 (2)

Q2 (5)

Hints and Solutions

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Q1 (2)

Given :

$$\vec{v} = 0.5t^2 \hat{i} + 3t \hat{j} + 9\hat{k}$$

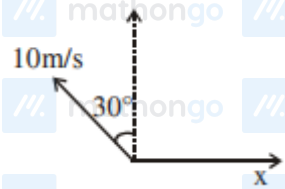
$$\vec{v}_{at=2} = 2\hat{i} + 6\hat{j} + 9\hat{k}$$

∴ Angle made by direction of motion of mosquito will be, $\cos^{-1} \frac{2}{11}$ (from x -axis) = $\tan^{-1} \frac{\sqrt{117}}{2}$

$$\cos^{-1} \frac{6}{11} \text{ (from } y \text{-axis)} = \tan^{-1} \frac{\sqrt{85}}{6}$$

$$\cos^{-1} \frac{9}{11} \text{ (from } z \text{-axis)} = \tan^{-1} \frac{\sqrt{40}}{9}$$

Q2 (5)



$$10 \sin 30^\circ = x$$

$$x = 5 \text{ m/s}$$